

Shelf Ready

Senior Full-Stack AI Engineer · Incubator Team

Five suppliers send us the same products described five different ways. Your job is to turn that mess into listings we could publish — **without publishing anything that isn't true**. This is the core loop our Incubator team ships, and the interesting part is not any single model call: it is deciding **which kind of work each step needs**.

What you're given

supplier_feed.json — 220 rows from 5 suppliers · **taxonomy.json** — 12 fixed categories

- The same product appears under **several suppliers**, with different titles, and sometimes **different numbers for the same spec**.
- Units, currencies and brand spellings are inconsistent (1.2kg vs 42 oz; SONY vs Sony Corp.).
- Some rows have **no specs at all**. Some have marketing shouting where the specs should be.
- Some rows are not products: test rows, an empty row, a mixed clearance pallet.
- Some titles are nearly identical but are **genuinely different products**.

```
{
  "row_id":      "row_9b67e4c70b",
  "supplier":    "NordicDist",
  "supplier_sku": "ND-4417",
  "raw_title":   "AeroBuds Pro True Wireless Earbuds - Black",
  "raw_specs":   "ANC; Bluetooth 5.3; 20h total playback with case; IPX4; USB-C",
  "price":       "£129.99",
  "stock":       184
}
```

Your task

- 1 **Match & normalise** — collapse the rows into **canonical products**: one entry per real product, with its supplier rows attached, and brands/units/prices normalised.
- 2 **Classify** — give each canonical product one category from **taxonomy.json**. The list is closed.
- 3 **Reconcile** — where suppliers disagree on a spec, decide what to publish *and* record that they disagreed.
- 4 **Write the listing** — generate a short customer-facing description for each canonical product.
- 5 **Verify before publishing** — check that **every claim in your description is supported by the source rows**. Anything unsupported must be withheld or flagged, never published.
- 6 **Route your uncertainty** — attach a confidence signal and put what you're unsure about into a human-review queue.
- 7 **Prove it works** — hand-label about **20 items**, then report both how well your matching agrees with your labels *and* how well your verification catches unsupported claims. Write a paragraph on where it fails.
- 8 **Show it** — one screen is enough: a canonical product, its source rows, the generated copy, and its flags.

The deliverable we care about most: LLM_ROLES.md

Several steps above want a model; at least one does not. We want to see **how you decide**. Give us one row per place a model is involved:

STEP	WHAT THE MODEL DOES	MODEL / TIER	WHY A MODEL, NOT CODE	WHEN IT'S WRONG, WHAT HAPPENS	ROUGH COST & LATENCY
e.g. matching	candidate grouping by meaning	embeddings, small	titles share no exact tokens	over-merge -> review queue	~1 call / row

And name at least one step where you deliberately did NOT use a model, and why. A pipeline that routes everything through one big prompt, and a pipeline that avoids models where they're clearly the right tool, are both wrong answers — we're interested in where you draw the line and how you justify it.

What to send back & how to work

DELIVERABLES

- A **Git repo** (link or zip) that runs from a clean clone, with a **README**: setup, how to run, how to run the eval.
- **LLM_ROLES.md** — the table above.
- A one-page **WRITEUP.md**: architecture, your 2–3 biggest trade-offs, what you'd do with another week, what you cut.
- An **AI_USAGE.md**: which AI tools you used, what you delegated vs wrote yourself, 2–3 example prompts, and one place the AI got it wrong and how you caught it.

HOW WE'D LIKE YOU TO WORK

- **Time-box: ~5–6 focused hours.** The pipeline and the roles table matter far more than UI polish. Skipping something and saying so is a signal, not a mark against you.
- **Any stack, any provider.** We'll give you an API key — **there is no cost or usage limit to worry about**, so choose models on merit, not on price.
- **Use your AI tools.** We build with Cursor / Claude / Copilot daily and expect you to.
- **Don't commit secrets.** Auth, deployment and multi-tenant are out of scope.

What happens next

A relaxed **30-minute call**: you walk us through the repo, we ask about a couple of your decisions — mostly from your roles table — and we'll ask you to make one small change live using your own tools. Nothing to prepare.

Reply with your repo link (or a zip) when you're done. Questions are welcome at any point — knowing when to ask a sharp one is part of the job.